(Please write your Enrollment No. immediately)

Enrollment No:.....

SUPPLEMENTARY MID-TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

Second Semester, March 2025

Paper Code: **BS106**

Subject: Applied Physics-II

Time: 1½Hrs.

Maximum Marks: 30

Note: Attempt Question no 1, which is compulsory, and any two more questions from the remaining

Q. No.	Question 1	Marks	CO
1(a)	Calculate the De-Broglie wavelength of 40KeV electron	[2.5]	CO1
1(b)	Find the average speed and root mean square speed for an assembly of two molecules, one with a speed of 1m/s and the other with a speed of 3m/s.	[2.5]	CO2
1(c)	Give physical interpretation of wave function ψ and its normalization.	[2.5]	CO1
1(d)	The Fermi level in potassium is 2.1 eV at a particular temperature. Calculate the number of free electrons per unit volume in potassium at the same temperature.	[2.5]	CO2
	Question 2		
2(a)	Set up the Schrodinger wave equation for a particle in an infinite well (1D). Obtain the wave functions and energy eigenvalues.	[5]	CO1
2(b)	What is Quantum Mechanical Tunneling	[2]	CO1
2(c)	An electron has a speed 500m/s accurate to 0.01%. With what fundamental accuracy can we locate the position of an electron?	[3]	CO1
	Question 3		
3(a)	What is Black Body Radiation? Derive Planck's formula for the energy distribution in the spectrum of a Black body.	[6]	CO2
3(b)	Fermi Energy in Ag is 5.51eV. What is the average energy of free electrons in Ag at 0K? What temperature is necessary for the average molecular energy in an ideal gas to have this value?	[4]	CO2
	Question 4		
4(a)	Compare the salient features of Maxwell Boltzmann(MB), Bose Einstein(BE) and Fermi Dirac(FD) Statistics	[5]	CO2
4(b)	Calculate the expectation value of the momentum of a particle trapped inside a 1D box.	[2]	CO1
4(c)	What is zero point energy in quantum mechanics? The energy of the linear Harmonic Oscillator in the third excited state is 0.1eV. Find the frequency of vibration.	[3]	CO1

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(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B. TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

2nd Semester, April-2024

Paper Code: BS-106 Subject: Applied Physics-II

Time: 1½ Hrs. Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question 1.	Marks	CO(s)
1(a)	Define Normalization of the wave function. If $\psi(x) = \frac{N}{x^2+a^2}$, calculate the normalization constant N.	2.5	1
1(b)	Show that if the uncertainty in the location of the particle is equal to its de-broglie wavelength, then the uncertainty in its velocity is equal to its velocity.	2.5	1
1(c)	Classify the following particles as either a Fermion or a Boson: (i) H^+, (ii) a photon, (iii) He^3, and (iv) an alpha particle. Give reason too.	2.5	2
1(d)	The Fermi level in potassium is 2.1eV at a particular temperature. Calculate the number of free electrons per unit volume in potassium at the same temperature.	2.5	2

Q. No.	Question 2.	Marks	CO(s)
2 (a)	Describe Davisson and Germer's experiment and explain how it enabled the verification of the wave nature of matter. What effect on the scattering angle in the Davisson-Germer experiment does increasing the electron energy have?	5	1
2 (b)	The eigen function for a particle in a 1-D box of length a is given by: $\Psi(x) = (2/a)^{1/2}\sin(m\pi x/a)$. Find the expectation value of position and momentum in the First excited state.	3	1
2 (c)	Discuss Zero Point energy of a Quantum Harmonic Oscillator.	2	1

Q. No.	Question 3.	Marks	CO(s)
3 (a)	What do you understand by Black Body radiations? Derive an expression for Planck's radiation law using Bose Einstein Statistics.	5	2
3 (b)	Prove that the average energy of an electron at T=0K is equal to $\frac{3}{5}E_F$, where E_F is the fermi energy at 0K.	3	2
3 (c)	Two particles are to be distributed in an energy level, which is 2-fold degenerate. Find the possible microstates if the particles are (i) distinguishable (ii) indistinguishable bosons (iii) indistinguishable fermions.	2	2

Q. No.	Question 4.	Marks	CO(s)
4 (a)	Show that the transmission probability for a particle of mass m and energy E in a rectangular potential barrier of height V_0 and width L is proportional to e^{-2kL}, where $k = \sqrt{\frac{8\pi^2 m(Vo-E)}{h^2}}$.	5	1
4 (b)	What do you understand by Death of a star? Comment on different stages.	3	2
4 (c)	Given Nitrogen Molecules N_2 at room Temperature 27 °C, calculate the average energy and most probable speed of molecules.	2	2

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(Please write your Exam Roll No.) Exam Roll No. .....

END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JUNE 2024

Paper Code: BS-106 **Subject: Applied Physics-II**

Time: 3 Hours **Maximum Marks: 60**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit. Assume missing data.

Q1 Answer the following questions: - **(4x5=20)**

a) Explain how the wave nature of particles give rise to uncertainty principle.

b) For an electron in one- dimensional box of width 2 A°, calculate the separation between the lowest two levels in eV.

c) A Bose-Einstein gas has two particles in the *i*th state whose degeneracy is three. Find the number of independent ways of selecting the particle in the state.

d) Draw sketch illustrating (011), (123), (111) and (001) planes in cubic unit cell.

e) What are Brillouin zones?

UNIT-I

Q2 a) What do you mean by the dual nature of matter and wave? Describe an experiment to support it. **(5)**

b) Set up Schrodinger equation for a free quantum particle. Discuss the properties of wave function. **(3)**

c) Calculate the expectation value $<p_x>$ of the momentum of a particle trapped inside a one-dimensional box. **(2)**

Q3 a) What is potential barrier and tunnel effect? Calculate the transmission probability for rectangular barrier for the condition of $E < V_0$, where E is the total energy of the particle. **(7)**

b) A particle constrained to move along the x-axis is described by the wave function

$\Psi(x) = 2x \quad 0<x<1$

$= 0$ elsewhere

Calculate the probability of finding the particle within the interval (0, 0 , 4). **(3)**

UNIT-II

Q4 a) Distinguish between quantum and classical statistics. **(3)**

b) How many photons are present in 100 cm^3 of radiation in thermal equilibrium at 1000 K.? **(3)**

c) Which type of statistics shall be applicable for a gas of (i) photons, (ii) electrons? Justify your answer. **(4)**

Q5 a) Show that the Fermi energy E_F of electrons in a metal at T = 0 is given by,

$$E_F = \frac{h^2}{2m}\left(\frac{3n}{8h}\right)^{\frac{2}{3}}$$

where symbols have their usual meanings. **(6)**

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b) Fermi energy for Gold is 5.54 eV. Calculate the Fermi temperature, given Boltz'man constant $1.38 \times 10^{-3} JK^{-1}$. **(2)**

c) How white/black dwarfs explain the concept of dying star? **(2)**

UNIT-III

Q6 a) Find the Miller indices of a plane that makes an intercept of 3A°, 4A° and 5A° on the coordinate axes of an orthorhombic crystal with a: b: c = 1: 2: 5. Find the equation of the plane. **(6)**

b) Using Bragg's equation, argue that greater is the angle of diffraction, greater is the accuracy in determining the lattice parameter. **(4)**

Q7 a) Differentiate the term amorphous and crystalline and amorphous solids. Write down seven crystal system with their lattice parameters. **(3)**

b) Show that the number of Frenkel defects in equilibrium at a given temperature is proportional to $(NNi)^{1/2}$ where N be number of atoms and Ni be the interstitial atoms. **(5)**

c) If X- rays of wavelength 0.5A° are diffracted at an angle of 5° in the first order, what is the spacing between the adjacent planes of the crystal? At what angle will second maximum occur? **(2)**

UNIT-IV

Q8 a) What are Bloch functions? Explain the origin of allowed and forbidden energy bands for electrons in solids. What is the number of orbitals in an energy bands? **(6)**

b) In an intrinsic semiconductor (Eg = 0.676 eV), m_e= 0.09 m and m_h= 0.36 m. Calculate the concentration of intrinsic charge careers at 300 K. **(2)**

c) What do you understand by effective mass of an electron? Explain its significance. **(2)**

Q9 Write short note on **(2x5=10)**

a) Zener Breakdown

b) Tunnel diode

c) LED

d) Fermi energy

e) PN-Junction diode

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END TERM EXAMINATION

SECOND SEMESTER [B.TECH] JULY 2023

Paper Code: BS-106 ***Subject: Applied Physics-II***

Time: 3 Hours ***Maximum Marks: 75***

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1 Answer the following questions:- **(3x5=15)**

(a) An eigen function of an operator $\frac{d^2}{dx^2}$, *is* $\Psi = e^{ax}$. Find the corresponding Eigen value.

(b) Show that Wien's law is a special case of Planck's radiation law.

(c) Lattice constant of a cubic lattice is 'a'. Calculate spacing between (011), (101), (112), (111) and (100) planes.

(d) Explain why semiconductor acts as insulator at 0K?

(e) The wave function of a particle is $\Psi = A\cos^2 x \; for \; interval - \pi/2$. Find the value of A.

UNIT-I

Q2 (a) What do you mean by particle in a box? Show that the energy of an electron in the box varied as the square of the natural numbers. **(3+7=10)**

(b) Prove quantum mechanically that particle will not exist in a box if its energy is zero. **(3)**

(c) Differentiate between $\Psi \; and \; |\Psi^2|$ **(2)**

Q3 (a) Distinguish between phase and group velocity. Show that the de-Broglie wave group associated with moving particle travels with same velocity as the particle. **(3+7=10)**

(b) What is the physical significance of wave function? **(3)**

(c) The particle trapped in one dimensional box of length L is described by a wave function Ψ= x. Normalise the wave function between a and b. **(2)**

UNIT-II

Q4 (a) State Planck's formula for Black body radiation and derive it from BE statistics. **(8)**

(b) If the Sun has a surface temperature of 5700K, what is the wavelength of maximum intensity of solar radiation? **(4)**

(c) Define Stefan's law. **(3)**

Q5 (a) An electron gas obeys the Maxwell-Boltzman statistics. Calculate average thermal energy (in eV) of an electron in the system at 300 K. **(4)**

(b) Distinguish between a Boson and Fermions. **(3)**

(c) What is the relative population of the first two single particle energy levels of a system of distinguishable particles if the energies of the levels is $\epsilon_0 = 0 \; and \; \epsilon_1 = kT$? **(4)**

(d) Discuss the Fermi-Direct distribution with the help of the distribution function; explain the concept of Fermi level and Fermi energy. **(4)**

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